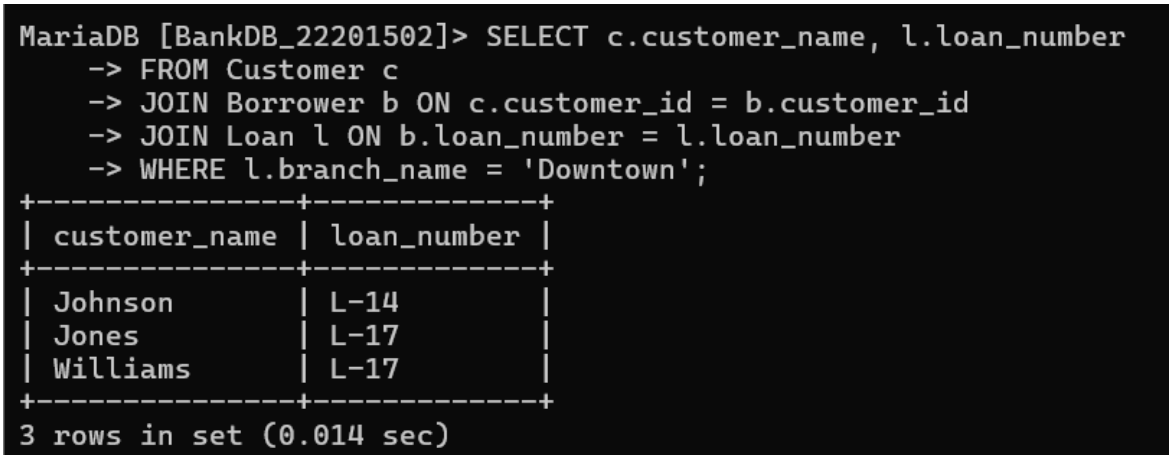
**CSE370 : Database Systems**

**Assignment 03 | Fall 2026**

**ID : 22201502 | Name : Abdullah al mahmud**

| | |
|---|---|
| **No 1 Query (as Plain Text)** | SELECT c.customer_name, l.loan_number<br>FROM Customer c<br>JOIN Borrower b ON c.customer_id = b.customer_id<br>JOIN Loan l ON b.loan_number = l.loan_number<br>WHERE l.branch_name = 'Downtown'; |
| **No 1 SS (of Query & Output in Shell)** |  |
| | |
| **No 2 Query (as Plain Text)** | SELECT DISTINCT c.customer_name<br>FROM Customer c<br>JOIN Borrower b ON c.customer_id = b.customer_id; |

**No 2 SS (of Query & Output in Shell)**

```
MariaDB [BankDB_22201502]> SELECT DISTINCT c.customer_name
    -> FROM Customer c
    -> JOIN Borrower b ON c.customer_id = b.customer_id;
+---------------+
| customer_name |
+---------------+
| Jones         |
| Smith         |
| Hayes         |
| Curry         |
| Williams      |
| Adams         |
| Johnson       |
+---------------+
7 rows in set (0.001 sec)
```

**No 3 Query (as Plain Text)**

```
SELECT c.customer_name
FROM Customer c
WHERE c.customer_id NOT IN (
  SELECT customer_id
  FROM Borrower
);
```

**No 3 SS (of Query & Output in Shell)**

```
MariaDB [BankDB_22201502]> SELECT c.customer_name
    -> FROM Customer c
    -> WHERE c.customer_id NOT IN (
    ->     SELECT customer_id
    ->     FROM Borrower
    -> );
+---------------+
| customer_name |
+---------------+
| Lindsay       |
| Turner        |
| Glenn         |
| Brooks        |
| Green         |
+---------------+
5 rows in set (0.013 sec)
```

**No 4 Query (as Plain Text)**

```
SELECT c1.customer_name AS Customer1,
     c2.customer_name AS Customer2,
     c1.customer_city AS City
FROM Customer c1
JOIN Customer c2
   ON c1.customer_city = c2.customer_city
  AND c1.customer_id < c2.customer_id;
```

**No 4 SS (of Query & Output in Shell)**

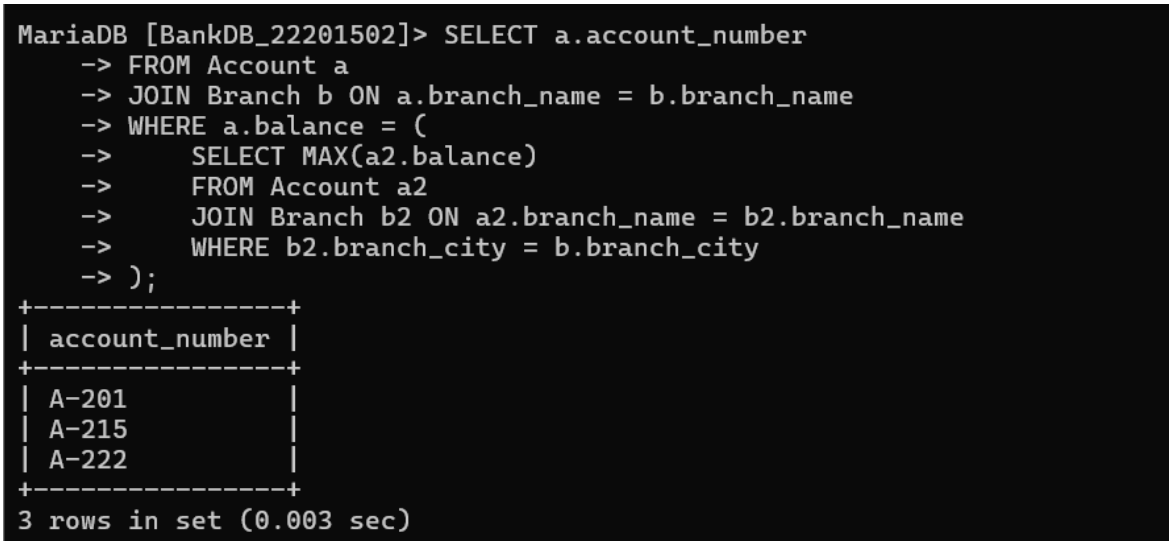
```
MariaDB [BankDB_22201502]> SELECT c1.customer_name AS Customer1,
    ->        c2.customer_name AS Customer2,
    ->        c1.customer_city AS City
    -> FROM Customer c1
    -> JOIN Customer c2
    ->     ON c1.customer_city = c2.customer_city
    ->    AND c1.customer_id < c2.customer_id;
+-----------+-----------+------------+
| Customer1 | Customer2 | City       |
+-----------+-----------+------------+
| Jones     | Hayes     | Harrison   |
| Smith     | Curry     | Rye        |
| Lindsay   | Adams     | Pittsfield |
| Turner    | Green     | Stamford   |
+-----------+-----------+------------+
4 rows in set (0.005 sec)
```

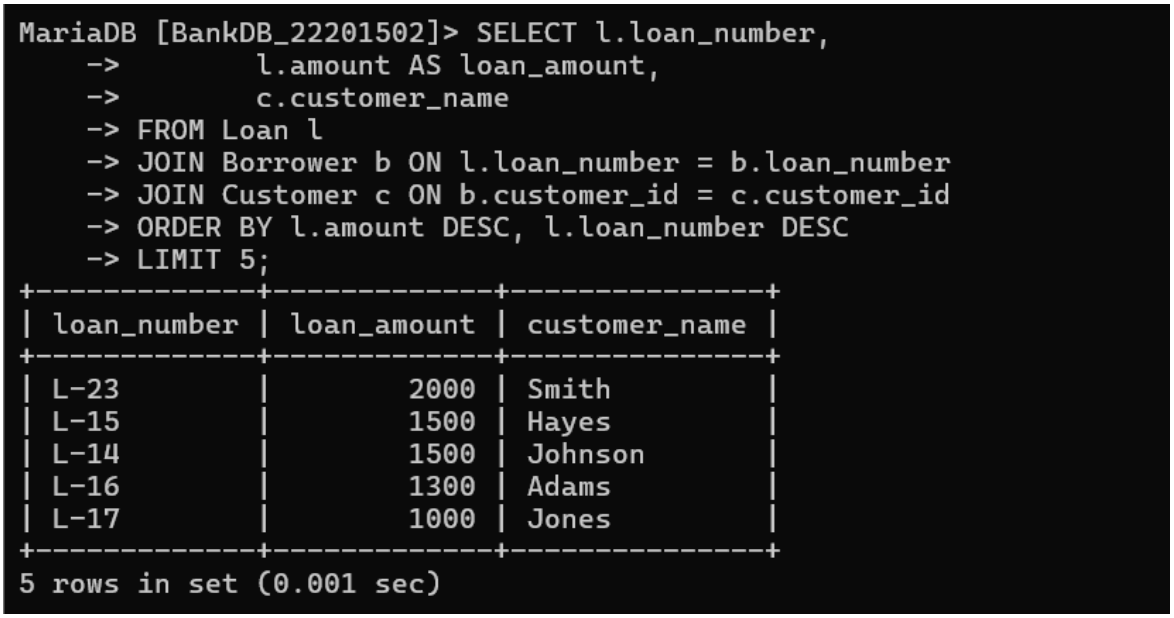
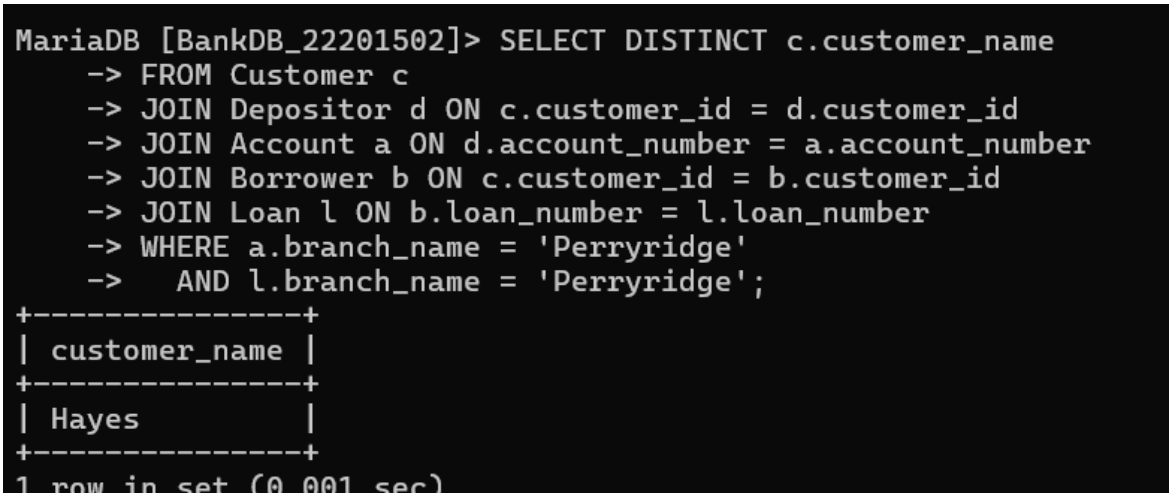
**No 5 Query (as Plain Text)**

```
SELECT branch_name AS Branch_name,
     ROUND(SUM(balance * 0.04), 2) AS Total_Interest
FROM Account
GROUP BY branch_name;
```

**No 5 SS (of Query & Output in Shell)**

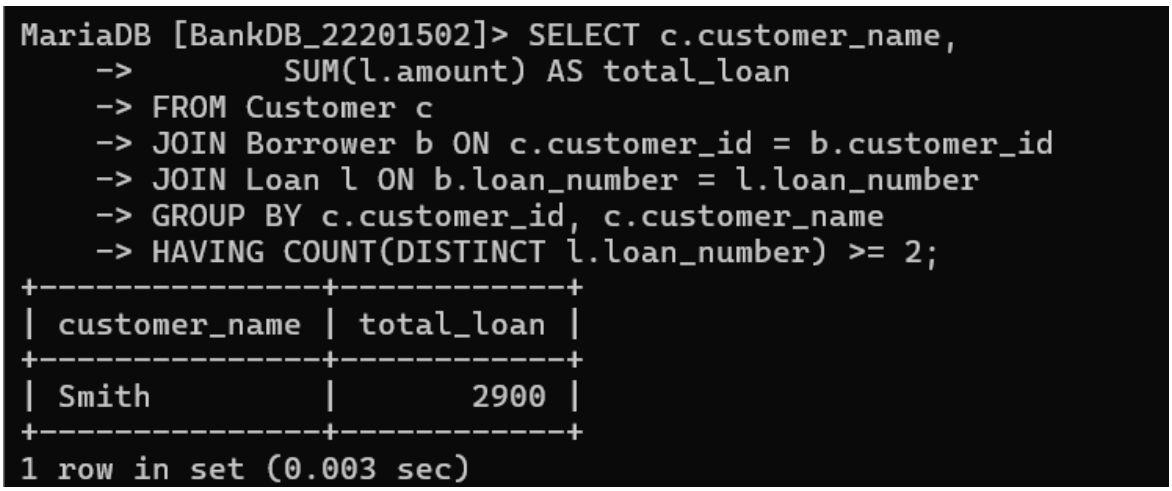
```
MariaDB [BankDB_22201502]> SELECT branch_name AS Branch_name,
    ->        ROUND(SUM(balance * 0.04), 2) AS Total_Interest
    -> FROM Account
    -> GROUP BY branch_name;
+-------------+----------------+
| Branch_name | Total_Interest |
+-------------+----------------+
| Brighton    |          66.00 |
| Downtown    |          20.00 |
| Mianus      |          28.00 |
| Perryridge  |          16.00 |
| Redwood     |          28.00 |
| Round Hill  |          14.00 |
+-------------+----------------+
6 rows in set (0.015 sec)
```

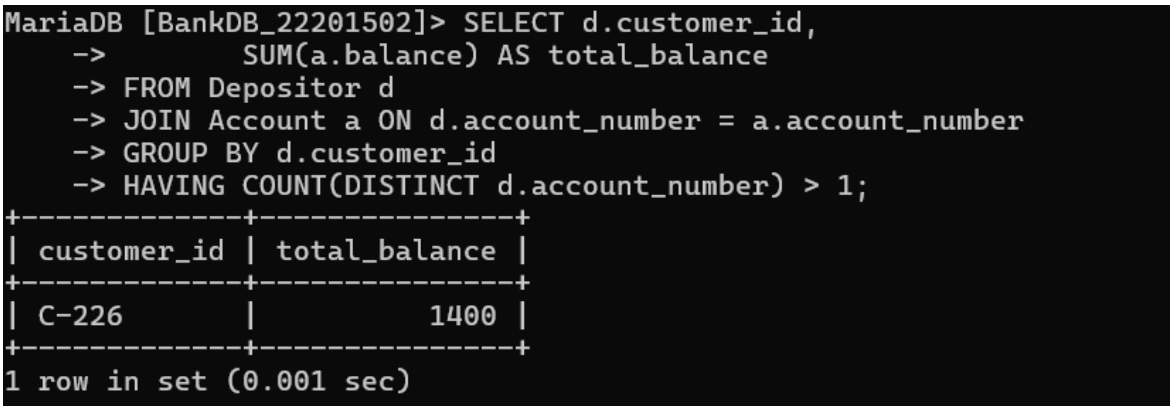
| | |
|---|---|
| | |
| **No 6 Query (as Plain Text)** | SELECT a.account_number<br>FROM Account a<br>JOIN Branch b ON a.branch_name = b.branch_name<br>WHERE a.balance = (<br>  SELECT MAX(a2.balance)<br>  FROM Account a2<br>  JOIN Branch b2 ON a2.branch_name = b2.branch_name<br>  WHERE b2.branch_city = b.branch_city<br>); |
| **No 6 SS (of Query & Output in Shell)** |  |
| | |
| **No 7 Query (as Plain Text)** | SELECT l.loan_number,<br>  l.amount AS loan_amount,<br>  c.customer_name<br>FROM Loan l<br>JOIN Borrower b ON l.loan_number = b.loan_number<br>JOIN Customer c ON b.customer_id = c.customer_id<br>ORDER BY l.amount DESC, l.loan_number DESC<br>LIMIT 5; |

| | |
|---|---|
| **No 7 SS (of Query & Output in Shell)** |  |
| | |
| **No 8 Query (as Plain Text)** | SELECT DISTINCT c.customer_name<br>FROM Customer c<br>JOIN Depositor d ON c.customer_id = d.customer_id<br>JOIN Account a ON d.account_number = a.account_number<br>JOIN Borrower b ON c.customer_id = b.customer_id<br>JOIN Loan l ON b.loan_number = l.loan_number<br>WHERE a.branch_name = 'Perryridge'<br>AND l.branch_name = 'Perryridge'; |
| **No 8 SS (of Query & Output in Shell)** |  |

```
MariaDB [BankDB_22201502]> SELECT l.loan_number,
    ->        l.amount AS loan_amount,
    ->        c.customer_name
    -> FROM Loan l
    -> JOIN Borrower b ON l.loan_number = b.loan_number
    -> JOIN Customer c ON b.customer_id = c.customer_id
    -> ORDER BY l.amount DESC, l.loan_number DESC
    -> LIMIT 5;
+-------------+-------------+---------------+
| loan_number | loan_amount | customer_name |
+-------------+-------------+---------------+
| L-23        |        2000 | Smith         |
| L-15        |        1500 | Hayes         |
| L-14        |        1500 | Johnson       |
| L-16        |        1300 | Adams         |
| L-17        |        1000 | Jones         |
+-------------+-------------+---------------+
5 rows in set (0.001 sec)
```

```
MariaDB [BankDB_22201502]> SELECT DISTINCT c.customer_name
    -> FROM Customer c
    -> JOIN Depositor d ON c.customer_id = d.customer_id
    -> JOIN Account a ON d.account_number = a.account_number
    -> JOIN Borrower b ON c.customer_id = b.customer_id
    -> JOIN Loan l ON b.loan_number = l.loan_number
    -> WHERE a.branch_name = 'Perryridge'
    ->   AND l.branch_name = 'Perryridge';
+---------------+
| customer_name |
+---------------+
| Hayes         |
+---------------+
1 row in set (0.001 sec)
```

| | |
|---|---|
| **No 9 Query (as Plain Text)** | SELECT c.customer_name,<br>SUM(l.amount) AS total_loan<br>FROM Customer c<br>JOIN Borrower b ON c.customer_id = b.customer_id<br>JOIN Loan l ON b.loan_number = l.loan_number<br>GROUP BY c.customer_id, c.customer_name<br>HAVING COUNT(DISTINCT l.loan_number) >= 2; |
| **No 9 SS (of Query & Output in Shell)** |  |
| | |
| **No 10 Query (as Plain Text)** | SELECT d.customer_id,<br>SUM(a.balance) AS total_balance<br>FROM Depositor d<br>JOIN Account a ON d.account_number = a.account_number<br>GROUP BY d.customer_id<br>HAVING COUNT(DISTINCT d.account_number) > 1; |

| | |
|---|---|
| **No 10 SS (of Query & Output in Shell)** |  |
| | |

```
MariaDB [BankDB_22201502]> SELECT d.customer_id,
    ->        SUM(a.balance) AS total_balance
    -> FROM Depositor d
    -> JOIN Account a ON d.account_number = a.account_number
    -> GROUP BY d.customer_id
    -> HAVING COUNT(DISTINCT d.account_number) > 1;
+-------------+---------------+
| customer_id | total_balance |
+-------------+---------------+
| C-226       |          1400 |
+-------------+---------------+
1 row in set (0.001 sec)
```